

Product Requirements Document (PRD)

SIH 26095 — AI-Driven Continuous Monitoring & Surprise Inspection System

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1. Problem Statement

Government schemes routed through NGOs, institutes, and funded projects are traditionally monitored through periodic, pre-announced inspections and self-reported paperwork. This creates two structural weaknesses:

Predictability — organizations know when inspections happen and can temporarily present a compliant picture (inflated attendance, staged activity, disabled CCTV).

Reactivity — irregularities (fund misuse, ghost beneficiaries, non-functioning projects) are typically discovered only after complaints or scheduled audits, long after the fact.

There is no system today that continuously ingests operational signals (attendance, CCTV, reports, complaints, feedback), computes a live risk score, and automatically triggers **unpredictable, risk-weighted, randomly assigned** field inspections — closing the loop from detection to verified evidence to government action.

2. Goal / Vision

Build a continuous monitoring platform that replaces scheduled, predictable audits with a closed-loop system:

Monitor → Detect → Score → Assign → Inspect → Verify → Act → Monitor again

The system should let a government authority (DoSJE or equivalent) register organizations, watch live risk indicators on a dashboard, and trust that high-risk institutes get surprise-inspected by an impartial, randomly/conflict-checked inspector, with all evidence tamper-evident and geo/time verified.

3. Target Users / Personas

Persona	Role	Needs
Government Official (DoSJE/Authority)	Oversees schemes, reviews risk dashboard, approves registrations, decides on corrective action	Real-time visibility, alerting, evidence-backed decisions

Persona	Role	Needs
Field Inspector	Receives assignments, travels to site, performs checklist-based inspection	Simple mobile app, offline capability, clear checklist, GPS-verified check-in
NGO / Institute Admin	Registers organization, uploads attendance/reports, responds to compliance notices	Simple registration/reporting flow, transparency on findings, ability to respond
Beneficiary (indirect)	May be contacted via random video verification	Minimal friction, no bias in selection
System Administrator	Manages platform, roles, integrations, security	Admin controls, audit logs, uptime

4. Scope Overview

The product is composed of six modules (as owned by the team):

M1 — Backend: core data model, workflow orchestration, APIs

M2 — AI/Risk Engine: anomaly detection, risk scoring, re-analysis

M3 — Mobile (Inspector App): assignment, navigation, GPS check-in, checklist, evidence capture, offline sync

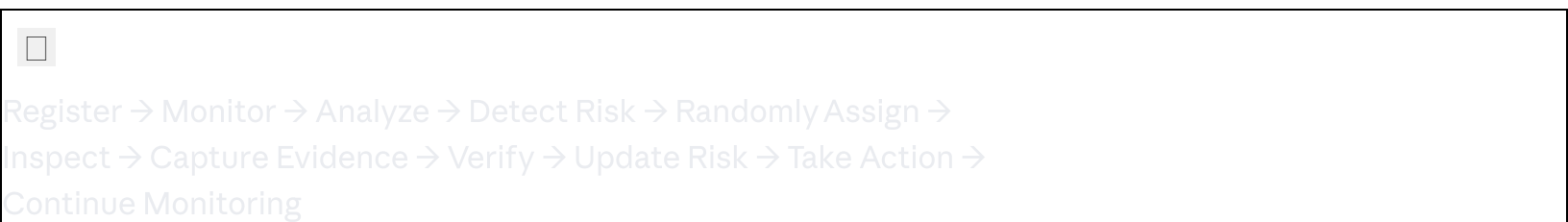
M4 — Web (Official Dashboard): registration approval, risk visibility, alerts, action management

M5 — Realtime: CCTV streaming/health monitoring, random video-conference verification

M6 — Security/DevOps: authentication, encryption, audit logging, deployment, integration security

5. End-to-End Workflow

5.1 High-level loop



5.2 Stage detail

Stage 1 — Registration NGO/Institute → Registration → Document Verification → Authority Approval → Project Activated. Data captured: organization details, institute/project, scheme, location, authorized staff, beneficiaries, documents, CCTV information.

Stage 2 — Continuous Data Collection Sources feeding the platform (not all continuously available):

Attendance (typically daily upload)

Project reports
CCTV (real-time where available)
Complaints
Beneficiary feedback

Stage 3 — Central Monitoring All sources (attendance, reports, CCTV, feedback, complaints, past inspections) flow into a Central Data Platform, which feeds the AI Engine. A government dashboard reflects current project status at all times.

Stage 4 — AI Anomaly Detection The AI engine flags signals without making accusations. Examples:

Attendance deviates far from historical average → attendance anomaly

Current report highly similar (~95%) to a previous report → possible reporting anomaly (copy-paste / fabricated reporting)

CCTV active hours drop sharply from baseline (e.g., 16 hrs/day → 2 hrs/day) → CCTV irregularity

Stage 5 — Risk Scoring (0-100) Weighted combination of signals, e.g.:

Signal	Weight
Attendance anomaly	+25
CCTV irregularity	+15
Previous violations	+20
Complaints	+10
Report inconsistency	+15
Inspection overdue	+10
Random factor	+5

Bands: **0-30 LOW · 31-60 MEDIUM · 61-80 HIGH · 81-100 CRITICAL**

Stage 6 — Alerting Dashboard surfaces high/critical projects with a reason list and two actions: *View Details / Initiate Inspection*. Inspections can be system-triggered per configured rules or manually initiated.

Stage 7 — Random/Risk-Based Assignment Assignment engine inputs: institute, risk, location, available inspectors, current workload, previous assignment history, conflict of interest (e.g., inspector previously inspected this same institute), specialization required, and a randomization factor — so organizations cannot predict or influence who inspects them.

Stage 8 — Mobile Notification Inspector receives a push notification with institute, location, priority, and deadline.

Stage 9 — Navigation & GPS Verification App provides map navigation. On arrival, GPS is checked against an allowed radius (e.g., 100 m). Outside radius → inspection is blocked/flagged; inside → inspection may proceed.

Stage 10 — Checklist-Driven Inspection Standard checklist, e.g.: institute operational?, staff present?, beneficiaries present?, activities being conducted?, required facilities available?, records available?, attendance correct?, CCTV functional?

Stage 11 — Evidence Capture Photos, videos, documents, notes, and attendance verification are captured. Every evidence item is stamped with GPS, timestamp, inspector ID, inspection ID, and a file hash (tamper-evidence). Flow: Mobile → Evidence Validation → Upload → Object Storage → Metadata written to DB → Audit Log entry.

Stage 12 — Offline Handling If connectivity is lost: save inspection and media locally, allow the inspector to continue and submit locally as "Pending Sync." On reconnect, a Sync Manager uploads evidence and submits the report, and the app confirms "Synced ✓."

Stage 13 — Inspection Report Inspector completes and submits the structured report (yes/no fields + free-text observations), e.g. flags where attendance doesn't match submitted records or CCTV is non-operational.

Stage 14 — AI Re-analysis The new field evidence is fed back into the AI engine, which recomputes risk using confirmed on-ground data (e.g., observed vs. reported beneficiary counts, confirmed CCTV outage, confirmed prior anomaly) — producing an updated, evidence-backed risk score.

Stage 15 — Government Action Official reviews AI alert + inspection report + geo-tagged evidence + historical data as one packet, then decides: issue notice, request clarification, corrective action, re-inspection, escalation, or close case.

Stage 16 — Compliance Tracking If corrective action is required: Official issues action → NGO is notified → NGO submits response → Official reviews → satisfied → Close Case; not satisfied → Re-inspection. This creates a full lifecycle rather than ending at the report.

5.3 Supporting Workflows

Random Video Conference (VC) Verification Official triggers "Random VC" → system selects an eligible institute/project and an eligible staff/beneficiary (not chosen by the institute) → secure room generated over WebRTC → call conducted → session logged → official records observations. Purpose: prevent institutes from presenting only a hand-picked "good" respondent.

CCTV Workflow Institute CCTV → RTSP/ONVIF → Media Gateway → WebRTC/HLS → Central Platform → Government Dashboard. In parallel, CCTV health is monitored; if a camera goes offline, this itself generates an anomaly signal into the risk engine — CCTV is a monitoring signal, not just a viewing feature.

5.4 Illustrative End-to-End Timeline

Time	Event	Risk
10:00 AM	Baseline status	42 (Normal)

Time	Event	Risk
11:30 AM	Attendance anomaly detected (reported 90 vs. historical avg 45)	42 → 67
12:00 PM	CCTV goes offline	67 → 78 (High)
12:15 PM	Inspection auto-triggered, inspector assigned	—
1:00 PM	Inspector arrives, GPS verified	—
1:05–1:30 PM	On-site inspection + evidence capture, report submitted (observed 42 vs. reported 90 beneficiaries)	—
1:35 PM	AI re-evaluates	78 → 94 (Critical)
1:40 PM	Official reviews full evidence packet, initiates corrective action	—

6. Functional Requirements by Module

M1 — Backend

- FR1.1: Organization/institute/project registration with document upload and multi-step approval workflow.
- FR1.2: Ingestion APIs for attendance, reports, CCTV metadata, complaints, and beneficiary feedback.
- FR1.3: Central data store unifying all source records per institute/project.
- FR1.4: Workflow orchestration for the full loop (register → monitor → ... → act).
- FR1.5: Role-based APIs for Official, Inspector, NGO Admin, and Admin personas.
- FR1.6: Audit log of all state transitions (registration approval, risk changes, assignments, actions).

M2 — AI / Risk Engine

- FR2.1: Attendance anomaly detection (deviation from historical baseline).
- FR2.2: Report similarity detection (duplicate/near-duplicate report flagging).
- FR2.3: CCTV activity anomaly detection (active-hours deviation, offline detection).
- FR2.4: Weighted risk scoring engine (0-100) combining all active signals, configurable weights.
- FR2.5: Risk banding (LOW/MEDIUM/HIGH/CRITICAL) and reason generation for alerts.
- FR2.6: Re-analysis pass incorporating inspection evidence to produce an updated risk score.
- FR2.7: Output must be advisory (flags/indicators) — never an automatic finding of wrongdoing.

M3 — Mobile (Inspector App)

- FR3.1: Push notification for assigned surprise inspections (institute, location, priority, deadline).
- FR3.2: In-app map navigation to institute location.
- FR3.3: GPS radius verification gating the start of an inspection.
- FR3.4: Digital checklist form matching the standard inspection template.

FR3.5: Evidence capture (photo, video, document, note, attendance verification) with GPS + timestamp + inspector ID + inspection ID + file hash on every item.

FR3.6: Full offline mode: local save of inspection + media, "Pending Sync" state, automatic sync manager on reconnect, "Synced ✓" confirmation.

FR3.7: Report submission with structured fields + free-text observations.

M4 — Web (Official Dashboard)

FR4.1: Organization/project registration review and approval.

FR4.2: Live project status view fed by the central platform.

FR4.3: Risk-ranked alert feed with reasons and one-click "Initiate Inspection."

FR4.4: Assignment visibility (which inspector, why, conflict-of-interest checks passed).

FR4.5: Evidence viewer per inspection (photos/video/docs with metadata).

FR4.6: Action panel: issue notice, request clarification, corrective action, re-inspection, escalation, close case.

FR4.7: Compliance-tracking view showing NGO responses and re-inspection outcomes.

FR4.8: "Random VC" trigger and eligible-participant selection UI.

M5 — Realtime

FR5.1: CCTV ingestion via RTSP/ONVIF, transcoded for web viewing (WebRTC/HLS).

FR5.2: CCTV health monitoring (camera-offline detection) feeding the risk engine.

FR5.3: WebRTC-based secure random video-conference sessions with session logging.

M6 — Security / DevOps

FR6.1: Role-based access control (Official, Inspector, NGO Admin, System Admin).

FR6.2: Encryption in transit and at rest for evidence, documents, and PII.

FR6.3: Tamper-evident storage (file hashing, immutable audit trail).

FR6.4: Secure, monitored deployment pipeline (CI/CD) across backend, AI, mobile, web, and realtime services.

FR6.5: Conflict-of-interest and randomization logic auditable and non-bypassable by any single actor.

7. Non-Functional Requirements

Reliability: risk scores and dashboards must reflect ingested data within a defined SLA (e.g., minutes, not hours).

Offline resilience: mobile inspection must be fully usable with zero connectivity and sync losslessly afterward.

Auditability: every risk change, assignment, and action must be traceable to its inputs (who/what/when).

Fairness/Non-predictability: assignment randomization and conflict-of-interest checks must be resistant to gaming by either institutes or inspectors.

Data privacy: beneficiary and staff personal data must be access-controlled and minimized.

Scalability: architecture should support many concurrent institutes, CCTV feeds, and inspectors nationally.

Usability: inspector app must be usable in low-connectivity, low-literacy field conditions.

8. Success Metrics

% of high/critical-risk projects inspected within target SLA (e.g., within 24 hours of risk crossing threshold).

Reduction in average time from anomaly detection to corrective action.

% of inspections with zero disputed/invalid GPS or evidence.

Inspector assignment fairness (distribution across inspectors, no institute repeatedly getting the same inspector).

Reduction in false positives flagged by AI over time (feedback loop from official decisions).

Compliance case closure rate and average cycle time.

9. Out of Scope (Full Product, not necessarily MVP)

Automated legal/financial penalty enforcement (system supports human decision-making only, per FR2.7).

Full national-scale CCTV integration across all camera vendors (initial support may be limited to common RTSP/ONVIF-compatible devices).

Deep NLP-based report content analysis beyond similarity/duplication detection (future phase).

10. Assumptions & Risks

Assumptions

Institutes have at least intermittent internet for attendance/report uploads.

CCTV, where present, is RTSP/ONVIF-compatible or can be bridged.

Inspectors are provisioned with a smartphone capable of running the mobile app.

Risks

CCTV vendor fragmentation could slow real-world integration.

Poor GPS accuracy in dense urban/indoor institute locations could cause false radius failures.

Over-reliance on AI risk scores without human review could create reputational/legal risk — mitigated by keeping AI output advisory only.

Connectivity gaps in rural areas require robust offline-first design on mobile (see FR3.6).

11. Team Ownership Mapping

Module	Owns
M1 Backend	Data model, workflow, APIs
M2 AI	Anomaly detection, risk scoring
M3 Mobile	Field inspection app

Module	Owns
M4 Web	Official dashboard
M5 Realtime	CCTV + video conferencing
M6 Security/DevOps	Pipeline security, deployment, integration